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DISRUPTIVE PERSPECTIVES

LETTER FROM SILICON VALLEY

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The purpose of Disruptive Perspectives:

When we analyze different themes, we spend a lot of time using many tools (quarterly reports, analyses, company dialogue, company visits, Excel, calculators and language models). We often make small notes, and sometimes larger notes that we think of as perspectives. We are old enough to know that there are rarely absolute truths, often only different perspectives.

Disruptive Perspectives has only one purpose: to share our perspectives on themes that shape our future. These are not academic papers, encyclopedia entries or recommendations to do, buy or sell anything. Just good old-fashioned information sharing to show how we view different themes at the time of publication. Perspectives do not become smaller when shared, perhaps rather larger. With that as our starting point, enjoy the journey through our perspectives.

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Disclaimer

The content of this article is not intended as investment advice or recommendations. If you have any questions about the funds referred to, you should contact a financial adviser who knows you and your situation. Remember also that historical returns in funds are never a guarantee of future returns. Future returns will depend on, among other things, market developments, the manager's skill, the fund's risk, and the costs of purchase, management and redemption. Returns may also be negative as a result of price losses.

We have recently been in Silicon Valley and met 20 companies. This travel letter is not a report from airports and meeting rooms. It is an attempt to give you the same thing we came home with, namely a sharper picture of the forces now shaping technology, capital and industry, and not least what that means for how we understand risk and positioning in the portfolio.



Silicon Valley is still the place where the next platform is built first, and where assumptions about the future are tested hardest. After 20 conversations in a short time, the individual pitches become less interesting than the pattern between them. What everyone takes for granted. What has suddenly become scarce. Where value is expected to end up as models become cheaper, energy more expensive and intelligence moves from the screen to physical work.

That is why the letter is built in two layers. First the big picture, that is, the underlying factors we believe are the foundation for both understanding and investing. Then the companies. Some meetings confirmed what we already believed. Others forced us to adjust the map. Both are valuable. We also include a discussion of some "new" key concepts that, during and after the trip, we found central to understanding, analyzing and investing.

Our goal is simple. That you should be able to read the world as we saw it over there, and understand why we believe certain themes deserve more attention than individual names. Over time, investments are rarely owned through the best story in the room. They are owned through exposure to the forces that allow many companies to be right at the same time.

Here is what we brought home.

Brief summary

1. Demand for AI infrastructure is still strong, and broader than a year ago. Practically all the suppliers we met are limited by what they can produce, not by what customers want to buy. The bottleneck is still supply, not demand.
2. The centre of gravity in AI investment has moved from training models to inference, that is, using them. Increased use changes which hardware is in demand, and which companies benefit.
3. Companies' own AI costs are growing fast. Budgets are increasing, partly at the expense of headcount growing more slowly or being reduced.
4. Open-source models for cost control and data security. The most sophisticated respond to rising costs by adopting open-source models on their own infrastructure.
5. Disaggregated and heterogeneous compute infrastructure is being implemented. The processing of an AI task is split up and distributed across tailored infrastructure. The concept is not new, but on this trip we saw concrete evidence that it is being implemented.
6. Optics and networking are becoming an ever more important part of AI infrastructure. Larger clusters, higher speeds and inference with low-latency requirements move more of the traffic onto light, and from 2028 also within the racks. The network increasingly determines how much you get out of the infrastructure.
7. Self-driving cars are taking over San Francisco. Tesla (Robotaxi), Google (Waymo) and Amazon (Zoox) are all on the roads in San Francisco. We tested Tesla to the airport (with a safety driver), and Waymo's new, larger, Chinese-made cars, while Zoox is still limited to test users.
8. AI is being adopted in every part of the value chain we met, from semiconductor production equipment to software, transport, banking and healthcare.

1.0 The big picture

After 20 meetings in Silicon Valley, you rarely come away with 20 equally important stories. You come away with a few big lines. That is the point of such a trip. Silicon Valley is not primarily a place. It is a concentrated laboratory where the infrastructure, business models

and capital structures of the future are tested in real time — often years before they reach the rest of the economy.

William Gibson has one of our favourite quotes:

«The future is already here - it's just not very evenly distributed»

Silicon Valley is where the future is unevenly, but most densely, distributed. When you meet 20 companies in a short time, you do not just see what they do. You see what they assume about the world, namely that the demand for intelligence is real, that energy, land and buildings ("land, power and shell") are now as decisive as the models, that AI is moving out of the screen and into physical work, and that the application layer (what Jensen Huang calls the top of the cake) is where value will ultimately be captured.

That is why the big picture is the foundation for both understanding and investing.

Understanding, because individual technologies without context become fashion. You can be fascinated by an agent, a humanoid or a new chip company and still miss that the real shift is a new industrial stack, such as energy, silicon, data centres, models, agents and then physical automation. Sequoia has put it this way: AI is not faster horses. It is cars. Then it is not just the product that changes. The whole value chain changes, the time horizon, and what counts as "edge".

Investing, because returns over time rarely come from owning the coolest story. They come from owning exposure to the underlying forces that allow many companies to be right at the same time, and from avoiding those that just ride the same narrative without a moat, without revenue or without access to the physical capacity the world actually lacks. Peter Thiel reminds us that "every moment in business happens only once". The next big position does not copy the last cycle. It captures what is unique now: that intelligence is scaling, that atoms matter again, and that capital follows the bottlenecks.

Marc Andreessen said that software is eating the world. In 2026 the picture is more precise: digital agents are eating the world, and the world is responding with power, chips, robots and new industry. The 20 meetings do not give the answer key. They give calibration. They show which assumptions have become consensus in the Valley, and therefore which questions an investor must ask harder. For example, who owns the lasting advantage when

models become cheaper? Who solves the physical constraint? Who builds something customers actually pay for, not just something that impresses in a demo?

The big picture is therefore not decoration on a travel report. It is the map. Without it, each company becomes an isolated bet. With it, the portfolio becomes an expression of how you believe the next decade will actually look.

Here are the lines we brought with us. And we are happy to share them, as long as you as a reader remember that we manage stocks, not truths.

1. Demand is strong and broad, and supply is the bottleneck

The most consistent message on the trip was that demand for AI infrastructure still exceeds production capacity. Nvidia was clear that revenue for the foreseeable future will be limited by access to components and production capacity, not by demand, and that any upside to next year's revenue must come from better deliveries from suppliers. Applied Materials has raised its growth estimate for equipment sales three times this year, to nearly 40%, and describes the availability of completed fabs as what governs growth in the short term. Arista ties its own growth directly to component allocations from Broadcom. Seagate is effectively sold out and lets customers commit to volumes several years ahead.

The bottlenecks sit in several places in the value chain. The most important are on the component side; TSMC, which manufactures the chips themselves for almost all the players, as well as suppliers of memory chips and optical components. Nvidia claims that access to energy is not an immediate bottleneck, but that it can cause deliveries to slip from one quarter to another.

One sentence from this meeting that stuck with us:

«The demand is not seen in technology before"

Demand is also broader than a year ago. It is no longer just the model providers driving demand. The large cloud providers (the hyperscalers) also supply businesses across industries. According to Nvidia, sales to governments and national AI initiatives tripled last year, and they expect this growth to continue. As an example, Jensen Huang mentioned at the Goldman Sachs conference (also in San Francisco) that Nvidia will deliver equipment for 2 GW of capacity to data centre operators in Australia, equivalent to USD 80 billion in 2027. Arista has more than a hundred customers building AI infrastructure, including universities

and industrial companies. NetApp, which sells storage systems to businesses, tripled the number of AI-related projects in one year.

A word that came up several times was "compute constrained", along with a sentence we took with us: *«we dont see any slowdown or pause in the horizon».*

2. From training to inference

An AI model is first trained on enormous amounts of data, and then put to use. In the first years of the AI wave, almost all investment went to training. Nvidia believes the tipping point was passed a year to a year and a half ago, and that the majority of what customers buy today goes to inference.

The drivers are so-called reasoning models, which use more processing power to work through a problem before answering, and AI agents, which carry out multi-step tasks on their own, as well as communicate with and delegate tasks to other agents. Both use far more tokens per query than a simple chatbot. Nvidia's assessment is that such breakthroughs increase the need for compute by a factor of 10 to 100, not by a few tens of percent, and that it therefore cannot be planned for within normal investment horizons.

Those who have read our notes know how we distinguish between linear and exponential models. We are living in the steepest part of the S-curve.

For us as investors, the point is that inference requires a different mix of hardware than training. It needs more fast memory close to the chips, more storage capacity to hold context in long tasks, more ordinary processors (CPUs) to orchestrate the agents, and more reliable networks. Nvidia's next platform, Vera Rubin, therefore includes dedicated rack types for CPUs, for low-latency inference and for storage, in addition to the classic GPU racks.

3. Tokens are the new currency, and the bill is growing

The work an AI model performs is measured in tokens (small pieces of text), and businesses pay per token. What was new on this trip was how large this cost item is becoming, and how aware companies are of it.

Some pointed out that tokens already make up around 10% of the cost base, and that the share is expected to keep rising. Headcount growth has been reduced to fund it, although they are still hiring in selected areas. At Applied Materials, the internal discussion is whether

they are spending enough on tokens given the return, while at the same time introducing consumption controls to avoid huge cost overruns. Lam Research said that some engineers have unlimited budgets, and that one day of agent-based work quickly costs around a thousand dollars. According to OpenAI, the most advanced enterprise customers use around 8 times as many tokens per employee as the average customer, and OpenAI itself is more than 30 times above that again. That gives an indication of how much growth lies in existing customers alone.

The pattern was the same almost everywhere. AI budgets are rising, headcount is growing more slowly, and productivity per employee is going up.

4. The most sophisticated use open-source models on their own infrastructure

When the token bill gets big enough, companies start looking for alternatives. The clearest example came from Nutanix. Their developers went from simple coding assistants to ever more powerful models, and costs to the providers of frontier models (the most advanced models from OpenAI, Anthropic and Google) rose sharply. The solution was to invest around USD 20 million in their own Nvidia servers and run open-source models on their own infrastructure. The goal is to cover around 80% of the need internally and send only the most demanding tasks to the frontier models. The investment is expected to pay for itself in 1 year, against a depreciation period of 5 years. Nutanix spoke of several large customers in semiconductors, pharma and finance considering the same.

Applied Materials wants to control which tasks are sent to open and closed models, based on cost, time and the need to protect its own technology. Arista pointed out that open-source models combined with data security requirements may lead companies to move AI work from the cloud back to their own data centres.

OpenAI, which lives off selling tokens, naturally had a counter-argument, namely that inference is not free for open-source models either, and they believe they are cheaper than the Chinese open models when comparing results. Whatever the outcome, both directions point to more demand for servers, storage and networking, whether it is open models on own infrastructure or open/closed models delivered by cloud providers.

5. Disaggregated inference and heterogeneous compute infrastructure are being adopted

In our view, this was the most important technical development on the trip. The concept has been discussed for a while, but this year we saw concrete evidence that it is being implemented.

The background is that inference consists of two phases with different hardware requirements. In the first phase (prefill), the model reads the entire question and all context, which requires a lot of raw compute. In the second phase (decode), the answer is generated token by token, which above all requires very high memory bandwidth (writing to/reading from memory). Nvidia's GPUs are well suited to the first phase. For the second phase there are specialized chips with memory integrated directly on the chip (SRAM), from companies such as d-Matrix, Groq (Nvidia) and Cerebras. Running the two phases on separate types of hardware is called disaggregated inference, and infrastructure that combines several types of chips is called heterogeneous.



Gimlet Labs is a cloud provider that specializes in this and develops software that routes work between the chip types. Their data show 4–10 times higher utilization on their infrastructure compared with pure GPU installations. d-Matrix said that disaggregation has expanded their market significantly, because their cards can now be used on the largest models together with GPUs, and that most customers choose heterogeneous setups because it extends the life of existing GPU fleets.

Confirmation came from several directions. Nvidia's own Vera Rubin platform is in practice disaggregated, with dedicated racks for CPUs, for low-latency token generation and for context storage. Nvidia's acquisition of Groq, which makes precisely SRAM-based inference

chips, points in the same direction. Other players also point to growing demand for CPU systems designed for agent and inference work. Although Nvidia has secured access to components and still appears well positioned, our conclusion is that value creation in AI infrastructure is spreading to more types of chips and more suppliers than a year ago.

6. Optics and networking are becoming ever more important

The AI infrastructure build-out is increasing demand for networking equipment. The network is increasingly decisive for how much customers get out of the chips, and a larger share of the traffic must move onto light. The industry divides the network into three levels. Scale-up is the connections between the chips in the same rack (or nearby racks), scale-out is the connections between racks in the same data centre, and scale-across is the connections between different data centres. We see three reasons why optics and networking are getting a larger share of investment at all three levels.

Scale-out networks barely existed before 2023 and have alone generated around ten times more optical traffic than the telecom era. The next step is scale-up. Copper is still both cheaper and more power-efficient than optics over short distances, but with each speed generation the distance at which copper is the best choice shrinks. When one coherent system has to span several racks, distance and speed requirements become too great, and Lumentum expects optics to be used in high volume within the scale-up domain for the first time in 2028, which is supported by Nvidia's plans. Once optics has to come in, power consumption argues for integrating it close to the chip (co-packaged optics, CPO) rather than in pluggable modules in the front panel, although some customers want to delay that transition as long as possible for reasons of maintenance and purchasing flexibility.

Training is steady, planned work. Inference for agents is unpredictable, with a lot of traffic back and forth between GPUs, CPUs and storage, and with low-latency requirements because the fastest tokens are priced highest. That makes synchronization and reliability more important. Credo showed how a single unstable link can halt an entire cluster of tens of thousands of GPUs, and that downtime costs millions per day. Reliability has thus become a purchasing criterion on a par with speed, also in the part of the network that connects the AI cluster to the rest of the data centre.

7. Self-driving cars are common in San Francisco

For those of us who visit San Francisco every year, self-driving cars are one of the clearest measures of how fast the technology is moving. This year we tested Tesla's robotaxi service several times, including to the airport. Tesla still has a safety driver in the seat, but the car drove itself the whole way. We saw significantly more Zoox cars than last year. Zoox is Amazon's self-driving car, built from the ground up without a steering wheel or pedals. Waymo, the market leader, has started rolling out its new car model, which is noticeably larger (and cheaper) than the Jaguar cars it has used until now, and which is made by China's Zeekr.



Nvidia described robotaxis as the first commercial breakthrough for physical AI, that is, AI that operates in the physical world, and confirmed that Waymo, Zoox and Tesla are all expanding to more US cities, which is also driving increased demand for AI infrastructure. They also claim that any carmaker that wants to match Tesla must multiply its purchases of training infrastructure.

We also met the two leading companies in self-driving trucks. Aurora has been driving without a driver on the highways of Texas since April 2025 and aims for 200 trucks on the road by the turn of the year. The service is priced below what a human driver costs per mile, and the truck can drive around twice as far per year because it is not subject to rest-time rules. Kodiak has 35 driverless trucks operating on private roads in Texas and aims for driverless operation on public highways by the end of 2026. California opened up for self-driving trucks this year, which expands the market significantly.

8. Edge AI: inference where the data is created

Not all inference will run in large data centres. A robot, a surveillance camera or a satellite cannot wait for an answer from the cloud. Latency becomes too long, power consumption too high, and the network connection is often unreliable. Inference that runs locally on the device is called edge AI.

Ambarella, which makes AI chips for cameras, cars and robots, was clear that data centre chips are not suited to such products; a robot cannot have a water-cooled GPU. By their own account, their chips use around a quarter of the power of a GPU with equivalent performance. They have delivered more than 50 million such chips and see robotics and drones as the fastest-growing part of the business. We also met Norway's Nordic Semiconductor, which, with its position in wireless communication, is working to put small AI functions directly into sensors and wearables. Planet Labs has put Nvidia chips on board its newest satellites, so that images are analyzed in orbit and only the result is sent down, for example the number of aircraft at an airport, instead of the whole image. Cisco envisions AI agents running locally on the office desk. The common denominator is that inference is moving to where the data is created, and that opens a market for other types of chips than those that dominate in data centres.

9. AI adoption across the value chain

We always ask companies how they use AI themselves. This year the answers were more concrete than ever, and they came from every link in the chain:

- Production equipment: Lam Research has trained robots (cobots) on the machines' own sensor data and maintenance manuals, so they can carry out repairs with a precision of 50 micrometres, consistently from shift to shift. Applied Materials uses AI to simulate experiments and discover new materials, and expects products to reach the market faster.
- Networking equipment: Cisco already ships products developed entirely by AI, and expects to have rewritten its entire historical code base with AI agents by next summer. With one of the newest models, they reviewed 11 billion lines of code in seven weeks, work that would normally have taken eight years. Integration with

Google took three weeks, compared with more than a year for similar partners previously.

- Chip design: OpenAI said that its own model was allowed to complete a list of known improvements to the company's first in-house chip in the last 30 days before the design was sent to production.
- Healthcare: Intuitive Surgical uses machine learning to warn the surgeon whether the tissue in the stapler is too thick or too thin before it fires. The path forward goes from the robot taking over individual tasks, such as camera control, to one surgeon supervising several operations at once.
- Banking: SoFi has launched an AI adviser in its app that, among other things, finds duplicate subscriptions and lets the customer cancel them with one click, and it has had more than half a million conversations since launch.
- Satellite data: Planet Labs won a contract with the US Navy because it photographs the entire South China Sea daily and lets AI pick out what the analysts should look at. By integrating AI tools on top of its data, it can also offer software that previously required customization for each segment, which both opens up the market and reduces development costs.
- Software development: Nutanix gets around 25 percent more features per developer per year. Arista has built an AI agent that monitors and troubleshoots the network

10. What we are watching closely

We also noted a few things that require attention. Energy is mentioned more and more often as the next bottleneck; gas turbines ordered today will not be delivered until 2030. Customer concentration is high at many suppliers, and continued investment growth will increasingly require debt financing. The hyperscalers describe memory as a cost driver in their own accounts, which is driving more innovation around new architectures that aim to reduce this cost (Nvidia is testing alternative versions of its next generation, Rubin Ultra, with fewer HBM layers). Geopolitics, the competitive situation and sales into and out of China continue to drive uncertainty across the value chain, including critical substrates such as indium phosphide (abbreviated "InP", used in optical components), semiconductor production equipment and memory (CXMT and YMTC). As we see it, multiples on many of these companies are not high overall (given expected earnings growth); in other words, the market is (to some extent) allowing for earnings levels not being sustainable. Applied Materials said

that questions about the duration of this investment wave have increased in recent quarters. Although uncertainty about where we are in the investment cycle is a recurring theme, the impression we came away with after the trip was that investment in AI is still in structural growth, and that we are still at an early stage of capturing the full value potential this technology makes possible.

2.0 The companies we met

After the big picture comes the evidence. The twenty companies are not a representative cross-section of Silicon Valley, nor are they a ranking. They are twenty conversations with people building ahead of what the market is already pricing in, and some building where the market is still hesitating.

Some sit in the middle of the model and agent layer. Some own the bottlenecks underneath: energy, chips, networking, data centres. Some take intelligence out into physical work, such as robotics, autonomy, manufacturing and defence. Some take the final step and turn the technology into something a customer actually pays for. Together they draw a map of where ambition, capital and scarcity meet right now.

We have not tried to reproduce every pitch. We have tried to capture what actually stayed with us after the meeting: what the company believes the world will be short of, why they think they can own part of the answer, and what it tells us about the moment we are in. Some meetings confirmed what we already suspected. Others adjusted the map. It is often the latter that are most valuable.

Each company is therefore presented with three things in mind: which line it belongs to, what sets it apart from the surrounding noise, and which question it leaves us with as investors. Not "is this the next unicorn?", but "if these theses are right, what must be true about the next five to ten years, and where should we stand?". Well, here are the 20.

AI chips and semiconductors

Nvidia (NVDA) – The world's largest supplier of AI chips (GPUs), and increasingly of complete data centre systems in which compute, networking and software are sold together. From the meeting: Revenue is limited by supply, not demand. The majority of purchases now go to inference. The next platform, Vera Rubin, includes dedicated racks for CPUs, low-latency token generation and storage. Nvidia is investing in optical suppliers to secure capacity. They

see robotaxis as the first commercial breakthrough for physical AI, and believe any carmaker that wants to match Tesla must multiply its purchases of training compute.

d-Matrix (private) – Semiconductor company designing chips tailored for inference. Its product Corsair is an accelerator card where compute and memory sit close together on the chip (in-memory compute based on SRAM), which makes the chip very fast in the decode phase. From the meeting: Has orders for the first half of 2027 and customers among neoclouds; hyperscalers and government players are in the pipeline. The next generation uses memory stacked directly on top of the chip.

Gimlet Labs (private) – San Francisco company that has built what it calls the first multi-silicon inference cloud. Its software splits a single AI request into phases and sends each phase to the chip that suits it best, whether GPUs, CPUs or specialized inference chips, and the company operates data centres built on this principle. The week before we met them, the company raised USD 300 million from, among others, Andreessen Horowitz at a valuation of three billion, and it says it has contracts worth several billion dollars and hundreds of megawatts. From the meeting: Has signed a large, multi-year agreement with a leading AI lab and works closely with Nvidia, Cerebras, d-Matrix and Google's TPU team.

Ambarella (AMBA) – Designs energy-efficient AI chips for cameras, cars, drones and robots, i.e. chips that interpret sensor data from the physical world. Has shipped more than 50 million such chips. From the meeting: The average price per chip has more than doubled in five years. The market estimate has been raised to over 20 percent annual growth. Robotics is growing fast, and a new stand-alone AI accelerator has been launched.

Rambus (RMBS) – Makes small but essential chips that sit between the processor and the memory on the server's memory modules. The chips are a small part of the module's cost, but the module does not work without them, and customers value supply security over price. Also sells chip designs (IP) and licenses patents. From the meeting: Chip sales grew around 40 percent last year. The next generation of processors gets more memory channels, which means more chips per server. Inference and agents increase the need for CPU capacity.

SiTime (SITM) – Makes timing chips, the reference clocks that keep all the components in a data centre synchronized, based on micromechanics (MEMS) instead of traditional quartz. From the meeting: Revenue has gone from just over USD 300 million last year to an annual run rate of more than a billion. Content per rack is increasing with denser racks, higher speeds and optics. Customer segments range from consumer electronics to data centre infrastructure.

QuickLogic (QUIK) – Small company making FPGAs, chips whose logic can be reprogrammed after manufacturing, for aerospace, defence and critical infrastructure. Also licenses the same technology into other companies' custom chips. From the meeting: Is developing, on behalf of the US Department of Defense, the first radiation-hardened FPGA for strategic weapon systems, with volume expected in 2028–2030.

Nordic Semiconductor (NOD) – Leading supplier of Bluetooth Low Energy, the technology that lets watches, headphones, sensors and smart-home products communicate wirelessly with minimal power consumption. The portfolio also includes other communication technologies such as cellular IoT and Wi-Fi, as well as power management chips, and in 2026 the company launched chips with built-in AI for local processing. From the meeting: Conversation with the company's US team about strategic customers, edge AI, the combination of compute and wireless communication, and memory shortages in the supply chain.

Networking and optics

Lumentum (LITE) – Makes lasers and optical components that convert electrical signals into light and back again, a prerequisite for thousands of AI chips to communicate. From the meeting: Growth is limited by its own production capacity, not demand. Has increased EML (electro-absorption-modulated laser, modules used in optical cables) capacity tenfold in two years and continues to expand across four fabs towards 2028, which it expects to be a year with a marked jump in demand as optics is adopted within the scale-up domain.



Arista Networks (ANET) – Makes Ethernet switches and the software that runs the networks in the world's largest data centres. From the meeting: AI revenue is expected to more than double this year, to at least USD 3.5 billion, with scale-across as a new growth engine. Customers are moving from Nvidia's InfiniBand to open Ethernet. Growth depends, among other things, on chip deliveries from Broadcom.

Cisco (CSCO) – The world's largest networking company, with security and collaboration tools (Webex, video equipment developed in Oslo) as well. From the meeting: An AI review of 11 billion lines of code is uncovering vulnerabilities that cannot be fixed in older hardware, which Cisco believes will force a replacement wave. Strong demand from governments for their own infrastructure.

Credo Technology (CRDO) – Specialist in high-speed connections: active copper cables that extend how far copper can reach, chips for optical modules and a new type of optics that warns before it fails. From the meeting: The optics business is guided to more than USD 600 million this year. Reliability has become a purchasing criterion because a single unstable link can halt an entire cluster of up to a million GPUs.

Napatech (NAPA, Oslo Børs) – Danish company supplying programmable network cards (SmartNICs) based on FPGA chips. The cards are inserted into servers and offload tasks that would otherwise take up much of the server's own processor capacity. Customers are providers of network monitoring and cybersecurity, telecom operators, governments and

financial players. The company has launched a 400 gigabit card aimed at AI data centres, with d-Matrix as its first customer.

Semiconductor production equipment

Lam Research (LRCX) – Makes the machines that etch and deposit layers on the silicon wafers in semiconductor fabs. The more three-dimensional the chip structures become, the more time the wafers spend in Lam's machines. From the meeting: Third year in a row of growth above the market for production equipment, and a record-high gross margin. Has started using robots for maintenance and repair of machines at customer sites, but it is still early days.

Applied Materials (AMAT) – The broadest equipment company in the industry, with around 19 percent of equipment investment in a typical fab, with tools for both advanced logic and DRAM memory. From the meeting: The growth estimate for equipment has been raised three times this year, to nearly 40 percent, and 2027 is expected to be strong. Access to completed fabs is what limits growth.

Memory, storage and infrastructure

Seagate Technology (STX) – One of two remaining hard disk manufacturers, which still store around 85 percent of the data in data centres. From the meeting: Now sells on orders with fixed contracts a year ahead and volume agreements several years out, with price set when the contract is signed. New laser-based technology (HAMR) gives more capacity per disk, and it plans no capacity expansions. Invests only 4–6 percent of revenue, so increased capacity comes entirely from technology improvements. Use of AI increases the need for storage. Rising sales to neoclouds, but still a small share compared with hyperscalers.

NetApp (NTAP) – Supplies storage systems to businesses, where the value lies more in the software than in the disks themselves. From the meeting: Product sales grew around 50 percent year on year last quarter, well above historical levels. The number of AI projects has tripled in one year, so far mostly data collection, but inference is expected to become the largest part. Has signed an agreement with Google for sovereign cloud solutions, probably the largest deal in the company's history. Three price increases this year due to higher component costs.

Super Micro Computer (SMCI) – Builds servers and complete racks for AI data centres, tested under full load at the factory so they can be put into operation the same day they arrive. From the meeting: Target of USD 68.5 billion in revenue in fiscal 2027, with 60 billion in the order book. Rising demand for CPUs; 10–20 percent of orders are CPU systems for agent work. Working to tighten payment terms and working capital after a few turbulent years.

Nutanix (NTNX) – Software that turns standard servers into a complete cloud platform, and the main alternative to VMware after Broadcom's acquisition. From the meeting: Bought its own GPU servers for USD 20 million and runs open-source models for its developers, with a one-year payback period. The servers customers buy have become four to five times more expensive in one year, which is delaying some projects. VMware migration is a 5–10 year journey; the migration process is largely automated, but AI can help customers with a lot of customization on top of VMware during the transition.

Self-driving vehicles and physical AI

Aurora Innovation (AUR) – Develops the Aurora Driver, a self-driving system for trucks, founded by entrepreneurs with backgrounds from Waymo and Tesla. From the meeting: Has been driving without a driver since April 2025 and aims for 200 trucks by the turn of the year. The price is below the cost of a human driver, the trucks can double their utilization, and more predictable driving requires less fuel. Partnerships with Volvo and PACCAR for trucks adapted to the system.



Kodiak AI (KDK) – Self-driving trucks without the need for detailed advance maps, with the same system for highways, industrial sites and defence. From the meeting: 35 driverless trucks operating around the clock in the Permian Basin in Texas, with 100 under contract. Aims for driverless operation on public highways by the end of 2026. Claims significantly lower capital consumption than its competitors.

Intuitive Surgical (ISRG) – The leader in robotic surgery with the da Vinci system, which was used in more than three million procedures last year. From the meeting: Believes 17 million procedures can still be moved over to robots. The fifth generation has force sensors in the instruments, a prerequisite for the robot eventually performing sub-tasks on its own. Remote surgery is expected in late 2027 or early 2028. Working with Nvidia on AI models.



Intuitive Surgical

QuantumScape (QS) – Develops solid-state lithium-metal batteries intended to deliver longer range, faster charging and better safety than today's lithium-ion. Licenses the technology rather than building factories itself. From the meeting: Works with four of the world's ten largest carmakers, with Volkswagen and Honda as the known ones. Also sees an opportunity in batteries placed by GPU racks in data centres from around 2028. Significant revenue is not expected until around 2029.

Software, platforms and data

SoFi Technologies (SOFI) – US digital bank that has gone from student loans to a complete financial platform with 15 million members. From the meeting: Revenue is growing 40 percent and earnings 60 percent. Half of all new products are sold to existing customers. Has launched an AI adviser and its own stablecoin.

Planet Labs (PL) – Develops and owns a fleet of small satellites and photographs the entire land mass of the Earth every day. From the meeting: New satellites with 30 cm resolution and a partnership with Nvidia to use GPUs on board to analyze images in orbit. Now also sells to governments (Japan, Sweden, Germany) that want to own the satellites. Defence and intelligence is the strongest driver. AI makes the image archive since 2017 more valuable, because they can offer the service across segments without building custom analysis tools for each segment.



Planet Labs

3.0 Some concepts that help you and me understand a little more of what is happening

Below are descriptions of words and concepts we discussed a lot during the trip. They increased our understanding; perhaps they will increase yours too? Heavy going, but probably worth it.

3.1 Heterogeneous data centres

Think of a data centre as a garage, and of inference as trips you need to make. The problem is not that you lack cars. The problem is that you have parked one type of car and are trying to use it for everything.

Inference is using a fully trained **model**. You ask a question, it answers **token** by token.

The job has two different phases:

1. Prefill = reading the whole question at once. Heavy, parallel, "the engine at full throttle".
2. Decode = writing the answer, one word at a time. Light for the engine, but you constantly have to retrieve what you have already said. It is a memory problem, not a horsepower problem.

GPU fill is how much of the car you actually use. A GPU can look "busy" while it is really waiting for the luggage. Then you are paying for a sports car idling outside the shop.

- **Homogeneous data centre** = a garage with only identical cars. Simple to own. A poor match when the trips are different.
- **Heterogeneous data centre** = a garage with several types of car, and someone who decides which car takes which leg.
- **One user vs. many**:One person waiting for the answer now, vs. a thousand jobs that just need to be done as cheaply as possible overnight.
- **Fastest vs. cheapest tokens**:How fast the text reaches you, vs. what it costs to produce a million such words.

The garage

You own a garage. The jobs look like this:

Trip	What it is	Wrong car	Right car
To the shop	Short, interactive, you wait	Truck	City car

Long highway trip	Long context, long answer	City car that must refuel every 10 min	Car with huge tank and speed
“Catch a flight”	One user, low latency	Scheduled bus	Sports car
Removals/night	Many users, high volume, price matters	Sports car	Truck

The cars in the garage

The truck: NVIDIA GPUs (xAI / Colossus style).

It can do everything. Training, prefill, decode, images, fine-tuning. You have many of them, they are standard, spare parts are everywhere.For a trip to the shop it is a waste. For removals it is unbeatable. When the garage is full of identical trucks and everyone just needs to “pop down to the shop”, the fill is poor. The engine is running, but the load bed is half empty.

Cerebras

The sports car with the fuel tank built into the chassis.

You do not have to stop and fetch fuel from an external depot for every token. That is why a single passenger can get text at 1,000–3,000 words per second.Expensive to buy, takes up a lot of space, and you cannot stack six different models in the same seat. Perfect when you are sitting and waiting. A poor choice if the job is hauling gravel all night.

d-Matrix

A van built to carry the goods in the seat, not in the back. Corsair can sit next to the truck in the same garage. It is strong when you need to write the answer fast without emptying the GPU’s memory bus. Raptor (the next model) promises an even shorter path between brain and luggage.Not a replacement for the truck. A specialist for the decode leg.

CPU / network

The bicycle courier. Fetches tools, searches the archive, opens a sandbox. Using the sports car for that is a waste.

Gimlet

Not a car. It is **the dispatcher** at the gate.You say “run this agent”. The dispatcher splits the trip: the truck reads the prompt, the sports car/van writes the answer, the bicycle courier fetches tools. Same garage, several cars, one trip. That is heterogeneous inference.

Postcards From The Future

Heterogeneous data centres are emerging – what are they?

Think of a data centre as a garage, and of inference as trips you need to make. The problem is not that you lack cars. The problem is that you have parked one type of car and are trying to use it for everything. Inference is using a fully trained model: you ask a question, it answers token by token.

Trip	**What it really is**	**Wrong car**	**Right car**
Down to the shop	Short, interactive, you wait at the door	Truck	City car
Long highway trip	Long context, long answer	City car that must refuel every 10 min	Car with huge tank and speed
“I have a flight to catch”	One user, low latency	Scheduled bus	Sports car
Removals/night delivery	Many users, high volume, price matters	Sports car	Truck

The truck: NVIDIA GPUs (xAI/Colossus style). The sports car with the tank built into the chassis: Cerebras. A van built to carry the goods in the seat, not in the back: d-Matrix	The bicycle courier: CPU / network. Fetches tools, searches the archive, opens a sandbox. **The dispatcher: Gimlet. Not a car. It is the dispatcher at the gate.**

Postcards From The Future

Heterogeneous data centres – many users and fast tokens

	One user (interactivity, TTFT, agents, voice, code)	**Many users** (aggregate tok/s, batch, overnight jobs, high occupancy)
Fastest tokens	Cerebras (and Groq-class LPU). 70B at ~2,100 tok/s, 120B class ~3,000 tok/s, trillion-MoE ~1,000 tok/s. Gimlet moves decode/spec-decode here and lets the GPU take prefill. d-Matrix Corsair as spec-decoder: 2–5× faster end-to-end vs. GPU-only spec-decode. Flagship Grok loses here.	Cerebras + Gimlet mix at high concurrent load, or dedicated decode racks. d-Matrix rack figures (30k tok/s on 70B) are aggregate, not per user. Homogeneous GPU farms can reach a high total, but each user sees 50–150 tok/s. xAI wins on capacity, not on tokens per second per session.
Cheapest tokens	Groq / small open models ($0.05–0.15/M in on 8B–20B). Cerebras is rarely cheapest per token — you pay for speed. xAI Fast ($0.20/$0.50) is "cheap enough" if you need Grok quality. d-Matrix's TCO promise is stronger when you own the rack than as a shared API.	Homogeneous GPU fill + d-Matrix TCO. When the batch is large, H100/B200/B300 win on $/token because the silicon is mass-market and already depreciated in training clusters (the Colossus logic). d-Matrix claims 3–13× tokens per dollar vs. A100/H100 on the 13B class in its own white papers — those are vendor figures.

- Fastest for one: You take the sports car (Cerebras/Groq class) for decode. You pay so that you do not have to wait.
- Cheapest for many: You fill the trucks (GPU halls like Colossus) and drive at night. The price per kilo falls when the load bed is full.
- Fastest for many: Either many sports cars (expensive), or the dispatcher who gives each leg the right car so the sum of trips goes fast without everyone sitting in the same queue.
- Cheapest for one: A small, cheap city car (small open models on Groq class / Grok Fast). Not flagship Grok in sports-car mode.

The common mistake is to buy only trucks because they won the last round (training), and then complain that the trip to the shop takes four minutes and costs as much as a house move. GPU fill is exactly that: the truck is "in use", but it is doing a job it was not built for.

In short: a homogeneous garage is simple. A heterogeneous garage is harder to own, but each trip costs less time or money, if the dispatcher knows which car should go out.

3.2 Scale up, scale out & across

The terms are used in several fields, but the core is the same: how you increase capacity. The difference lies in what you make bigger – one unit, more identical units, or more places and contexts. This has become the new tribal language of data centres. Nobody fully agrees on the definitions, but below is our understanding. The next quarterly reports will be hard to follow without understanding the difference between them. Or so we believe.

Brief overview

Term	Core idea	Typical question
Scale up	Make the unit stronger	Can we give this machine, team or method more power?
Scale out	Add more identical units	Can we spread load over more nodes, teams or places?
Scale across	Connect places, systems or contexts	Can several data centres, markets or organizations act as one system?

The three are not mutually exclusive. In modern AI infrastructure they are often used at the same time.

Scale up – vertical scaling

Scale up means increasing capacity inside one existing unit. In IT this is called vertical scaling. More CPU, RAM, GPU, memory bandwidth or faster internal communication in the same server.

IBM describes it like this: *scale-up is done "by adding more resources to an existing system to reach a desired state of performance".*

In AI data centres, scale-up often means packing more GPUs into the same node or rack and connecting them tightly (NVLink, NVSwitch and similar), so that they act as one large computer with shared memory and very low latency.

Strengths: Easy to understand. Good for workloads that need shared memory and tight coordination. Low internal latency.

Limitations: A physical ceiling. Chassis, power, cooling and bus width. One large node becomes a single point of failure. The cost per performance gain often rises steeply.

You upgrade the road surface so that the same road can handle more traffic – instead of adding more lanes.

Scale out – horizontal scaling

Scale out means increasing capacity by adding more nodes and spreading the work between them. IBM ties this to distributed architectures and "pay-as-you-grow".

In practice: more servers behind a load balancer, more pods in Kubernetes, more shards in a database, more GPU nodes in a training cluster. Each node is an independent compute unit; the network ties them into a resource pool.

Strengths: Almost unlimited growth in theory. Better fault tolerance (one node can drop out). Often cheaper commodity hardware. Suits stateless services, microservices and data-parallel training.

Some limitations are that it requires splitting the workload. More complexity, such as orchestration, consistency, networking and observability. Communication between nodes costs latency and bandwidth. When communication takes more time than computation, the system stops scaling linearly ("communication bound").

In other words, you add more lanes so that more cars can drive in parallel.

Hyperscalers (Google, Amazon, Microsoft) are largely built on scale-out: many relatively identical machines, not one supercomputer. Britannica sums up the distinction like this: traditional data centres *"scale up" by upgrading existing hardware; hyperscalers "scale out" by adding more computers.*

Scale across – across places and systems

Scale across is the newest and least standardized term. In the AI and networking industry from around 2025–2026, it has taken on a fairly precise meaning:

Scale-out usually stays within one data centre. Scale-across connects resources in different geographic locations – campus, metro, region or cloud – via DCI, optical transport or WAN, so that several clusters can act as one logical system.

Drivenets puts it briefly:

«Extending high-performance networking across multiple data centers to support workloads too large for a single site»

Ciena describes the point as the ability to train one AI model on GPUs that are physically located in different campuses or regions – "one logical AI training environment, even when the infrastructure is physically distributed".

NVIDIA calls scale-across "*a new dimension, orthogonal to the existing connectivity options of scale-up and scale-out".*

Why did the term appear now? One data centre hits a ceiling on power, land and cooling. The largest models require more energy than one site can deliver. Clusters then have to span several buildings.

The price: Higher latency, more jitter, lower effective bandwidth, more expensive optics. Synchronous training over long distances is demanding; that is why scale-across is more often used for task distribution, redundancy, inference and "near-synchronous" training than for tight GPU-to-GPU communication.

A quote that shows the definition is still in motion:

«Scale-up is highly localized. It's synchronous, message passing type, low latency... Scale-out is basically within the pod. ... Scale-across can vary [depending] on who you talk to. It can be 10 kilometers to thousands of kilometers.»

How they fit together

In modern AI factories, the three are layers, not alternatives:

1. Scale up makes each node as fast as possible (intra-node / intra-rack).

2. Scale out ties the nodes into a cluster in one data centre.

3. Scale across ties the clusters into a larger, geographically distributed system.

A typical comparison:

Dimension	Scale up	Scale out	Scale across
Scope	One node / rack	Many servers, one DC	Several DCs/regions
Goal	Max perf. per unit	Large cluster	Global / multi-site
Distance	Under ~10 m	10 m–2 km	10 to 1,000s of km
Network	NVLink, NVSwitch, CXL	InfiniBand, RoCE, Ethernet	DCI, optics, WAN
Latency	Lowest	Medium	Highest
Scalability	Ltd. by chassis/power	High within a DC	High reach, costly

Sources for the table: Optcore, Asteraix, HYTOPTODEVICE.

A practical rule of thumb: do you need more power in one coherent computation (shared memory, low latency)? → Scale up.

Do you need more total capacity and robustness, and can the work be split? → Scale out.

Has one location, one system or one context hit a physical, political or organizational ceiling? → Scale across.

Most systems that become truly large use all three: first more muscle in each unit, then more units, then more places acting as one.

3.3 Light, mirrors & optics

Lumentum and Coherent do not sell “AI chips”. They sell light sources and optical building blocks that let GPUs, TPUs and switches talk to each other when copper cables become too slow, too hot and too short. That is why demand is so strong, and why 2028 is often called a turning point. We call it “The Optical Big Bang happens in 2028”.

The simple idea first

Think of an AI cluster as a city.

- Copper is the streets inside one block. Works well over short distances.
- Optics is the motorways. Light carries more data, further, with less loss.
- As clusters grow and speeds go from 800G to 1.6T and 3.2T, the motorway has to be pulled into the block itself – first between racks, then inside the rack, and finally right up against the chip itself.

Lumentum and Coherent make the “light bulbs”, the modulators, the modules and increasingly entire optical systems that make that move possible.

Concepts & tribal language

Postcards From The Future

Glossary of the tribal language of optics

InP (indium phosphide)
The semiconductor material most high-speed lasers are made of. Hard to produce, small wafers, long qualification. That is why InP capacity is the real bottleneck.

EML (electro-absorption modulated laser)
A laser that both creates the light and encodes the data on the same chip. The workhorse in many 800G and 1.6T modules. Lumentum has a particularly strong position here.

CW laser (continuous wave)
A laser that only creates a stable, continuous light. The data encoding happens in another component, often silicon photonics. Needed in SiPh modules and in CPO/NPO.

UHP laser (ultra-high-power)
Extra powerful CW laser. One laser can feed many channels. The key to external light sources for CPO/NPO.

ELS (external laser source)
The laser sits outside the hot switch/XPU package and sends light in via fibre. Easier to service than gluing it onto the ASIC.

Pluggable module
Today’s standard: a box you plug into the front of the switch (800G, 1.6T, later 3.2T).

NPO (near-packaged optics)
The optics sit right next to the chip on the board. Simpler than CPO, faster to introduce, often used as a bridge to full CPO.

CPO (co-packaged optics)
The optics sit in the same package as the switch or XPU chip. Shorter electrical path, lower power and lower latency – but harder to make, test and replace.

OCS (optical circuit switch)
An optical “cross-connect” that passes light on without first converting it to electricity. Used to reconfigure AI networks, save power and route around bad nodes.

Postcards From The Future

Four ways to scale AI, with different optical intensity

Term	What it means	Distance	Optics need
Scale-across	Between data centres/campuses	Long	Pump lasers, DCI, ZR+
Scale-out	Between racks in the same hall	Medium	800G/1.6T pluggables, eventually CPO switches
Scale-up	Inside the cluster/between GPUs	Short	Copper today, NPO/CPO from 2027–2028
Scale-in	Inside the package itself	Very short	Future optics close to the XPU

Lumentum's Wupen Yuen summed up the bandwidth like this:
scale-across = 1, scale-out = 10, scale-up = 100, scale-in = 1,000.

Why 2028 is called a "big bang"

It is not one product that explodes. It is several curves meeting. They converge:

1. Next-generation XPUs and faster SerDes arrive. CPO/NPO is attached to new platforms, not yesterday's GPUs.
2. Optical scale-up starts in late 2027 and is set to grow in 2028. NPO may arrive a quarter before CPO.
3. 3.2T modules go from early volume in 2027 to a large share in 2028. Goldman, among others, has sharply raised its estimate for the entire global module market towards 2028.
4. The copper limit moves inward in the rack. The first scale-up optics also span several racks, not just within one.
5. New laser capacity arrives late: Coherent doubles InP again in 2027; Lumentum's Greensboro fab is expected to generate revenue in early 2028. The demand is there before full qualified capacity.

Lumentum, on the turning point itself, or the Big Bang

«Lumentum expects NPO deployments to begin around late 2027 and identifies 2028 as a key industry inflection point for optical Scale Up.»

CEO Michael Hurlston, on demand and bookings:

«The capex numbers from the US hyperscalers are enormous and there seems to be no end in sight. We're falling further and further behind the demand. We would be sold out through all of 2028 within two quarters.»

Hurlston, on why the architecture is shifting:

«Lumentum is positioned at the heart of a secular industry shift. As AI compute workloads increase in both speed and bandwidth, data center architects are turning to optical links as a primary means of connectivity.»

Lumentum on the timing for scale-up lasers:

«We remain confident that demand for ultrahigh-power laser chips will begin ramping in the second half of 2027 in preparation for customers' scale-up deployments in 2028.»

Wupen Yuen on how big it could become as optics goes deeper in:

«Scale across is one, scale out is 10 in bandwidth, scale up is 100, scale in is 1,000 … as intensity of optical demand increases, and with XPU numbers going up, I would say Lumentum's best days are yet to come.»

Coherent CEO Jim Anderson on visibility:

«Customer demand remains exceptionally strong with no signs of attenuation, and our visibility continues to extend further into the future with orders now reaching into calendar 2028 and customer LTAs extending to the end of the decade.»

Anderson on CPO as a market:

«CPO is … a transformational growth opportunity for the company. We see that market size as over $15 billion, and that's probably a conservative estimate over the coming years. … scale-up CPO revenue in the second half of calendar '27.»

Anderson on the thesis itself:

«AI scales on optical connectivity.»

This is how it fits together in one sentence. Or so we believe.

2026–27 is a boom in pluggable 800G/1.6T and in the laser chips inside them.2028 is the year optics is expected to leap into the AI cluster (NPO/CPO scale-up), while 3.2T and OCS scale up – and InP lasers are still scarce.

That is why it is not "just another transceiver cycle". It is a shift from copper + pluggable box to light as infrastructure inside the machine. Lumentum and Coherent hold the light source that transition requires.

4.0 Closing thoughts

We are lucky to be able to travel around Silicon Valley during what we believe will be called the craziest period on the planet so far this millennium. Many thanks to DNB Carnegie, Christoffer and Mona Cathrin, who arranged yet another wonderful trip.

A trip to Silicon Valley does not give you the answer key. It gives you calibration.

We did not come home with twenty investment recommendations and a finished map. We came home with a sharper distinction between what has become consensus, what is still underestimated, and what is just noise dressed in good language. The big picture is what we take with us into our ongoing work: how we read companies, which questions we ask, and where we believe risk and return actually lie in the years ahead.

Unitholders' and investors' capital should not chase the most impressive pitch. It should own exposure to forces deep enough to carry several companies - and be disciplined enough to say no when the story is strong but the assumptions are thin. That is the standard we bring back from Silicon Valley.

Thank you for your trust. We continue the work.

See you in the future.



This travel letter is based on meetings in Silicon Valley on 7–11 September 2026 and reflects the companies' own statements and our impressions. It is not intended as investment advice.

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